

MC: Multivariate Calculus - Integration Techniques:

⇒ Basic integration Rules:-

- substitution Methode
- Integration by Part:-

→ substitution for integrals it is used when when it's (any exp) derivative's exist
using substitution we can solve complex problems easily.
ex,

$$\int (3x^2+7) (x^3+7x)^8 dx$$

this expression derivative exist
Hence we will substitute it.

Let $\boxed{x^3+7x=t}$

Now find the derivative of assumption:-

$$\Rightarrow (3x^2+7) = \frac{dt}{dx}$$

$$(3x^2+7) dx = dt$$

→ Now we have to replace $(3x^2+7) dx$ with dt

→ So

$$\begin{aligned} \textcircled{1} \Rightarrow \int t^8 dt &= \frac{t^{8+1}}{8+1} + c \\ &= \frac{t^9}{9} + c = \frac{1}{9} t^9 + c \end{aligned}$$

so now replace t .

$$\frac{1}{9} (n^3 + 7n)^9 + C$$

Ans:-

ex:- $\int (n^2 - 2) e^{n^3 - 6n + 4} dn$

let: $n^3 - 6n + 4 = t$

$$3n^2 - 6 = \frac{dt}{dn}$$

$$dn (3n^2 - 6) = dt$$

Now divid $\frac{1}{3}$ both side.

$$\frac{1}{3} (3n^2 - 6) = \frac{1}{3} dt$$

$$\boxed{n^2 - 2 = \frac{1}{3} dt}$$

Now Replace

$$\int \frac{1}{3} dt \cdot e^t = \frac{1}{3} \int e^t dt \rightarrow \text{Now solve the int.}$$

$$e^t \text{ integration} = \int e^n dn = e^n + C$$

so $\frac{1}{3} e^t + C$

Now replace t

$$\frac{1}{3} e^{n^3 - 6n + 4} + C$$

Note: we find substitution method only the case where its derivative should exist.